Biomedical Text Embeddings Weakly Capture Chromatin Accessibility Similarity: A Large-Scale Measurement Study

***Abstract*—The extent to which biomedical text metadata can serve as a proxy for chromatin accessibility similarity remains unknown. We conducted a systematic measurement study comparing pairwise cosine similarity matrices across 11 distinct embedding types (Atacformer, regulatory potential (RP), four biomedical language models, term frequency-inverse document frequency (TF-IDF), and a random Gaussian baseline) across 32,245 human assay for transposase-accessible chromatin with sequencing (ATAC-seq) experiments from ChIP-Atlas. Using Spearman rank correlation, Mantel permutation tests, and top-k neighborhood overlap, we find that text embeddings of experimental metadata show weak but statistically significant correlation with Atacformer-derived chromatin similarity ($r = 0.06$ at cell-type level; $r = 0.25$ to 0.28 at full scale). TF-IDF matches or outperforms all large language models (LLMs), indicating that surface-level vocabulary rather than semantic understanding drives this correlation. Biomedical language models marginally outperform a general-purpose encoder, but neither approach overcomes the TF-IDF baseline. Text embeddings share only 2 to 3% of nearest neighbors with Atacformer but 20 to 23% with regulatory potential, revealing that text captures cell-type-level clustering recoverable from metadata but not the richer chromatin landscape encoded by Atacformer. These results establish quantitative bounds on the utility of text metadata as a proxy for chromatin similarity and motivate continued development of chromatin-specific representation models.**

***Index Terms*—genomics, bioinformatics, chromatin, natural language processing, transformers, feature extraction, correlation, similarity measures**

I. Introduction

The rapid growth of epigenomics databases such as ChIP-Atlas [1], Encyclopedia of DNA Elements (ENCODE) [2], and Roadmap Epigenomics [3] has produced tens of thousands of ATAC-seq experiments across diverse human cell types and conditions. Organizing, retrieving, and comparing these experiments requires similarity measures that reflect the underlying biology. A natural question arises: to what extent can text metadata associated with experiments (cell type labels, experimental descriptions, and publication titles) serve as a proxy for chromatin accessibility similarity?

This question has direct practical implications. If text embeddings of experimental metadata closely approximate chromatin-based similarity measures (such as those derived from Atacformer or RP), they could enable natural-language retrieval and annotation of ATAC-seq data without access to the raw signal, potentially democratizing the use of large epigenomics repositories. If agreement is limited, the results instead point to fundamental limits on what text can capture and motivate investment in chromatin-specific representation learning.

Recent advances in biomedical natural language processing have produced powerful text encoders pretrained on large biomedical corpora. PubMedBERT [4] is pretrained from scratch on PubMed abstracts and full-text articles. SapBERT [5] and MedCPT [6] are both initialized from a pretrained PubMedBERT checkpoint and further fine-tuned for different objectives: SapBERT via self-alignment on Unified Medical Language System (UMLS) synonym pairs, and MedCPT via contrastive learning on PubMed search logs. These models achieve strong performance on biomedical entity recognition, relation extraction, and information retrieval, yet whether their semantic representations capture the biological relationships encoded in chromatin accessibility data has not been systematically evaluated.

Atacformer [7] is a transformer trained on ATAC-seq data using triplet margin loss to produce experiment-level embeddings that cluster by cell type and chromatin state. Regulatory potential (RP) [8] provides a complementary, functional-genomics-derived signal: a gene-activation-weighted summary of chromatin accessibility indexed by gene transcription start sites (TSS).

We compare pairwise cosine similarity matrices from 11 embedding types (chromatin models, biomedical LLMs, TF-IDF, and a random baseline) across 32,245 curated human ATAC-seq experiments, treating both agreement and disagreement as informative findings.

The key findings are:

- Text is a weak chromatin proxy: vocabulary overlap alone (TF-IDF) matches or exceeds all biomedical LLMs, indicating semantic understanding adds little beyond surface-level word matching.
- Text captures coarse cell-type identity but not fine-grained chromatin variation.
- These bounds establish a ceiling on text-based epigenomics retrieval and motivate chromatin-specific representation learning.

II. Methods

Fig. 1 summarizes the overall study design: chromatin-based embeddings (Atacformer, RP) and text-based embeddings (biomedical and general language models, TF-IDF, random) are computed for each experiment, converted into pairwise similarity matrices at two levels of aggregation, and compared using the three metrics described in Section II-E.

A. Dataset

We obtained 48,822 human ATAC-seq experiments from ChIP-Atlas (hg38 assembly) [1]. We removed 16,567 experiments with corrupted or empty BED files and 10 experiments producing zero regulatory potential signal near gene TSSs, yielding a curated set of 32,245 experiments spanning 1,108 cell types.

B. Atacformer Embeddings

We encoded each experiment using databio/atacformer-base-hg38 (AtacformerForCellClustering) [7], a transformer trained with triplet margin loss. Peak BED files were tokenized using the gtars Tokenizer with a vocabulary of 890K genomic regions. Up to 1,024 randomly sampled peak tokens were mean-pooled to produce a 192-dimensional embedding per experiment.

C. Regulatory Potential Embeddings

We computed RP for each experiment using a LISA2-style exponential decay model [8]: each chromatin peak's contribution to a given gene is weighted by $w(d) = e^{-d/10\,\mathrm{kb}}$, where d is the distance to the gene's TSS, with exon-adjacent peaks receiving additional upweighting. Summing across all peaks yields a per-gene RP score; stacking over all hg38 autosomal gene TSSs maps each experiment to a 40,772-dimensional vector.

D. Text Embeddings

This section describes text- and metadata-based embedding methods only; chromatin-based embeddings (Atacformer and RP) are described in the two preceding subsections.

For each experiment we constructed a full metadata string by concatenating available fields: {cell_type}. {cell_type_class}. {cell_type_description}. {title}. We embedded this string using four encoders, all of which produce 768-dimensional embeddings: PubMedBERT [4] (microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext, mean pooling), SapBERT [5] (cambridgeltl/SapBERT-from-PubMedBERT-fulltext, classification token (CLS) pooling), MedCPT [6] (ncbi/MedCPT-Query-Encoder, mean pooling), and a General encoder [10] (sentence-transformers/all-mpnet-base-v2, mean pooling). To isolate the contribution of cell type identity, we also encoded each experiment using only the cell type name string with all four models (cell-type-name-only ablation). We additionally computed TF-IDF embeddings (top 2,048 features, sublinear TF, 2,048-dimensional) and a random Gaussian baseline (192-d, fixed seed).

E. Similarity Matrices and Comparison Metrics

For each embedding type we constructed pairwise cosine similarity matrices at two levels of aggregation.

Cell-type-aggregated level. Individual experiment embeddings were mean-pooled within each cell type to produce one representative vector per cell type. A corpus mean vector was computed as the arithmetic mean over up to 10,000 randomly sampled individual experiment embeddings and subtracted from each cell-type mean vector. For every pair of cell types (i, j), the cosine similarity between their centered vectors was computed and stored in entry (i, j) of the matrix, yielding a symmetric $1{,}108 \times 1{,}108$ matrix with ones on the diagonal.

Full-dataset level. The same corpus-centering procedure was applied to all 32,245 individual experiment embeddings. For every pair of experiments (i, j), the cosine similarity between their centered embedding vectors was computed and stored in entry (i, j), yielding a symmetric $32{,}245 \times 32{,}245$ matrix with ones on the diagonal.

Corpus centering corrects for anisotropy in transformer mean-pool representations: without it, embeddings tend to cluster in a narrow cone in the embedding space, which inflates cosine similarities uniformly and obscures relative differences in pairwise structure. The centering vector was estimated from individual experiments (not from cell-type means) to avoid conflating aggregation level with the centering baseline.

We quantified agreement between each text-embedding similarity matrix and each chromatin reference (Atacformer and RP) using four metrics:

1) Spearman rank correlation (r) on upper-triangle edge weights.
2) Mantel permutation test (1,000 permutations) for empirical p-values at the cell-type level.
3) Bootstrap confidence intervals for r (1,000 resamples of cell types/experiments, with 50,000 randomly sampled pairs per resample), reported at both the cell-type and full-dataset scale as a scale-independent measure of effect-size precision.
4) Top-k neighborhood overlap: mean fraction of shared top-k nearest neighbors, $k \in \{5, 10, 20, 50\}$.

All analyses were conducted with both Atacformer and RP as reference structures (i.e., the chromatin similarity structures against which text-based embeddings are evaluated). Mantel tests were GPU-accelerated on an NVIDIA GB10 GPU ($\sim$30 s per condition vs. $\sim$38 min on CPU).

F. Decomposing LLM Signal from TF-IDF

To determine whether LLMs encode biological information beyond surface-level vocabulary overlap, we conducted three complementary analyses comparing the full-text LLM embeddings against TF-IDF at both the cell-type-aggregated and full 32,245-experiment levels.

Pairwise similarity scatter. For each LLM, we sampled up to 25,000 random experiment pairs and plotted TF-IDF pairwise similarity against LLM pairwise similarity, coloring each point by Atacformer similarity (Fig. 5). Points above the diagonal

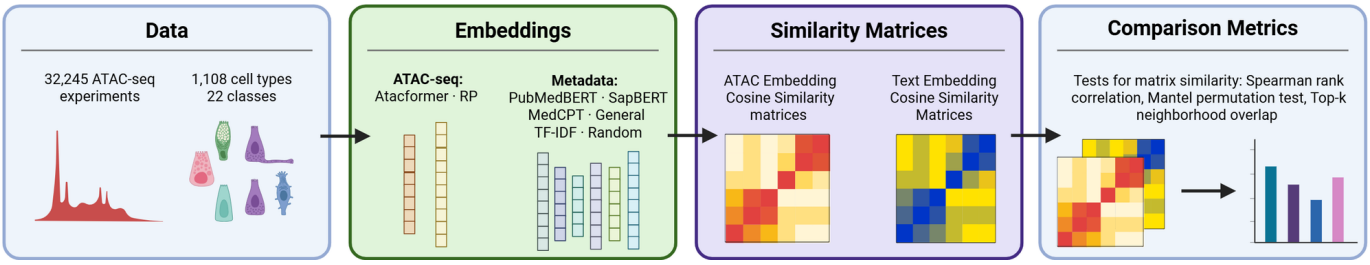


Fig. 1. Study design. ATAC-seq experiments are embedded using chromatin-based models (Atacformer, RP), text encoders (biomedical and general language models, TF-IDF), and a random baseline. Pairwise cosine similarity matrices are computed at cell-type-aggregated and full-dataset levels and compared using three complementary metrics. Created with BioRender.com [9].

indicate pairs where the LLM assigns higher similarity than TF-IDF; points below indicate the reverse. The Spearman correlation between TF-IDF and LLM similarity vectors was computed over the same sample.

Partial Spearman correlation. We computed the partial Spearman rank correlation of each LLM similarity matrix with Atacformer, controlling for TF-IDF:

$$r_{LM,AC|TF} = \frac{r_{LM,AC} - r_{LM,TF}\, r_{AC,TF}}{\sqrt{(1 - r_{LM,TF}^2)(1 - r_{AC,TF}^2)}} \qquad (1)$$

where $r_{X,Y}$ denotes the Spearman correlation of the upper-triangle edge weights of matrices X and Y. At the full scale, correlations were estimated on a random subsample of 2×10^6 pairs from the $\sim$520M-element upper triangle. A positive partial correlation indicates that the LLM encodes biological structure beyond vocabulary overlap; a value near zero or negative indicates its Atacformer alignment is fully explained by TF-IDF (i.e., once TF-IDF's shared variance with Atacformer is removed, the LLM contributes no additional association with chromatin similarity).

Cross top-k neighborhood overlap. For each LLM and each $k \in \{5, 10, 20, 50\}$, we computed the mean fraction of top-k LLM nearest neighbors that are also top-k TF-IDF nearest neighbors, using the same topk_overlap procedure described above. High overlap indicates the LLM's local structure recapitulates TF-IDF; low overlap indicates genuinely distinct neighbor assignments.

III. Results

A. Text Embeddings Show Weak but Significant Alignment with Atacformer

Sanity checks confirmed that Atacformer and RP produce biologically coherent similarity structures, with immune, neural, and cancer cell types forming distinct clusters, while text-based embeddings show coarser organization (Fig. 2).

At the cell-type-aggregated level, all four full-text biomedical language model embeddings showed weak but statistically significant Spearman rank correlation with Atacformer (PubMedBERT: $r = 0.060$; SapBERT: $r = 0.057$; MedCPT: $r = 0.035$; General: $r = 0.032$; all Mantel $p < 0.01$, $N = 1{,}000$ permutations; Table I). TF-IDF showed comparable agreement ($r = 0.060$, $p < 0.001$). Cell-type-name-only variants of the same models showed no significant correlation with Atacformer ($r \approx 0$, $p > 0.20$ for all four models), establishing that agreement in full-text models is driven by experimental description and title metadata, not cell type name alone.

TABLE I

Spearman rank correlation (r) between each embedding type and Atacformer similarity, at the cell-type-aggregated level ($N = 1{,}108$ cell types) and the full-dataset scale ($N = 32{,}245$ experiments, $\sim$520M pairs), with 95% bootstrap confidence intervals (1,000 resamples; see Methods). Significance markers (Mantel permutation test) are reported at the cell-type level only; at full scale, p-values are uninformative and omitted.

Embedding	r (cell-type)	95% CI	r (full)	95% CI
RP	0.315^{***}	$[0.284, 0.357]$	0.216	$[0.206, 0.227]$
PubMedBERT (full text)	0.060^{***}	$[0.034, 0.098]$	0.276	$[0.265, 0.287]$
SapBERT (full text)	0.057^{***}	$[0.036, 0.088]$	0.248	$[0.237, 0.258]$
MedCPT (full text)	0.035^{**}	$[0.016, 0.067]$	0.229	$[0.219, 0.239]$
General (full text)	0.032^{***}	$[0.017, 0.060]$	0.196	$[0.186, 0.206]$
TF-IDF	0.060^{***}	$[0.035, 0.098]$	0.300	$[0.290, 0.311]$
PubMedBERT (name only)	-0.006^{ns}	$[-0.018, 0.017]$	0.090	$[0.080, 0.100]$
SapBERT (name only)	0.003^{ns}	$[-0.014, 0.029]$	0.137	$[0.127, 0.147]$
MedCPT (name only)	0.004^{ns}	$[-0.012, 0.032]$	0.068	$[0.059, 0.078]$
General (name only)	0.005^{ns}	$[-0.009, 0.030]$	0.090	$[0.081, 0.099]$
Random Gaussian	-0.000^{ns}	$[-0.004, 0.015]$	0.000	$[-0.009, 0.009]$

$^{***}p < 0.001$; $^{**}p < 0.01$; ^{ns} not significant ($p > 0.05$). Mantel permutation test.

At the full 32,245-experiment scale, global rank correlations (i.e., across all $\sim$520M experiment pairs) were higher for all models. TF-IDF achieved the highest unadjusted Spearman correlation ($r = 0.300$), outperforming all LLMs including PubMedBERT ($r = 0.276$), SapBERT ($r = 0.248$), MedCPT ($r = 0.229$), and the general-purpose encoder ($r = 0.196$; Table I).

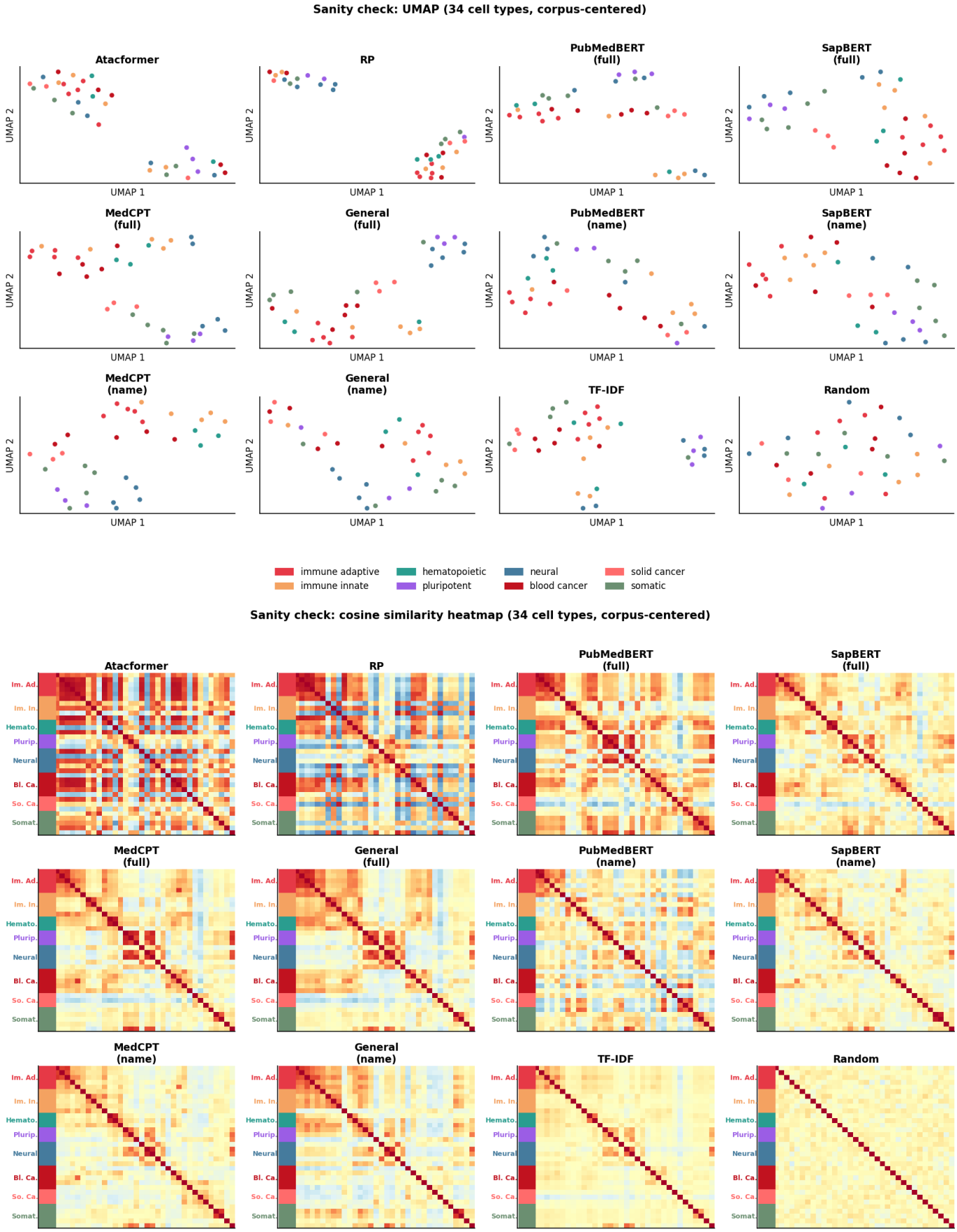


Fig. 2. Sanity check Uniform Manifold Approximation and Projection (UMAP) projections (top) and cosine similarity heatmaps (bottom) for all 12 embedding conditions over 34 representative cell types. Atacformer and RP show biologically coherent lineage clustering; text-based models show coarser, cell-type-level organization.

B. Biomedical Language Models Encode Modest Signal Beyond Vocabulary Overlap

To determine whether the observed text-Atacformer correlation reflects information beyond vocabulary overlap, we computed the partial Spearman correlation (see Section II-E) of each LLM with Atacformer after controlling for TF-IDF (i.e., removing variance in LLM-Atacformer similarity attributable to vocabulary overlap; Fig. 3).

PubMedBERT ($\Delta r = +0.051$) and SapBERT ($\Delta r = +0.045$) retained small but non-zero positive partial correlations after controlling for TF-IDF, indicating a residual association with Atacformer similarity that is not explained by vocabulary overlap alone. MedCPT ($\Delta r = -0.030$) and the general-purpose encoder ($\Delta r = -0.042$) showed negative partial correlations, meaning their unadjusted Atacformer agreement is fully (and nominally more than) accounted for by shared vocabulary with TF-IDF. The high correlation between LLM and TF-IDF similarity structures ($r = 0.73$ to 0.82) confirms that vocabulary overlap is the dominant driver across all models.

PubMedBERT, SapBERT, and MedCPT are all derived from the same PubMedBERT-pretrained base (Section 2.4), so this divide cannot be attributed to differences in base biomedical-literature pretraining alone. PubMedBERT (no further fine-tuning) and SapBERT (fine-tuned via UMLS synonym self-alignment, an objective that preserves general semantic-similarity structure) both retain a positive residual, whereas MedCPT (fine-tuned via contrastive learning on PubMed search-click logs, an objective oriented toward query-document retrieval relevance rather than topical similarity) does not. We emphasize that a positive partial correlation only establishes that the residual association is non-zero; it does not by itself establish that this residual reflects biologically meaningful semantic understanding.

This pattern is also visible in the pairwise similarity scatter (Fig. 5): for PubMedBERT and SapBERT, experiment pairs scored higher by the LLM than by TF-IDF (above-diagonal points) carry warmer Atacformer similarity values than below-diagonal pairs (Fig. 6). This asymmetry is weaker for MedCPT and the general encoder, consistent with the partial correlation results above.

Cross top-k neighborhood overlap between LLM and TF-IDF neighbor lists was $\sim$33% at $k = 5$ across all four models (Fig. 4), indicating that roughly two-thirds of each LLM's local structure is genuinely distinct from TF-IDF. However, the partial correlation results above show that this unique structure aligns with Atacformer only for PubMedBERT and SapBERT.

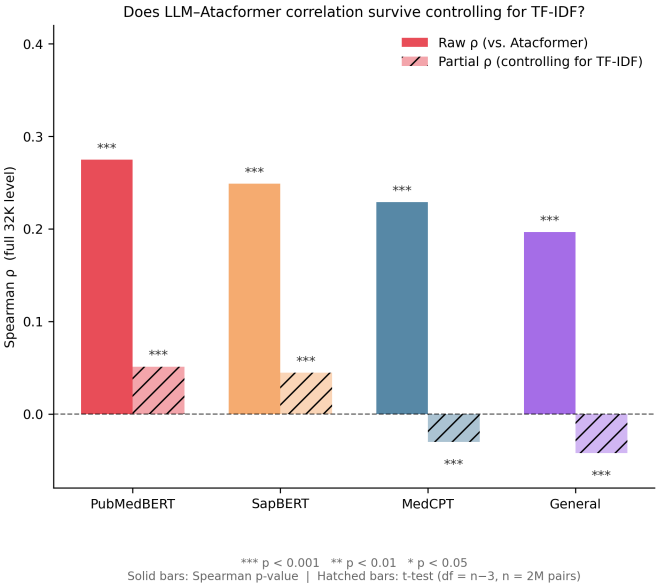


Fig. 3. Spearman r vs. Atacformer (solid bars) and partial Spearman r after controlling for TF-IDF (hatched bars), full 32,245-experiment level. PubMedBERT and SapBERT retain positive partial correlations; MedCPT and General do not.

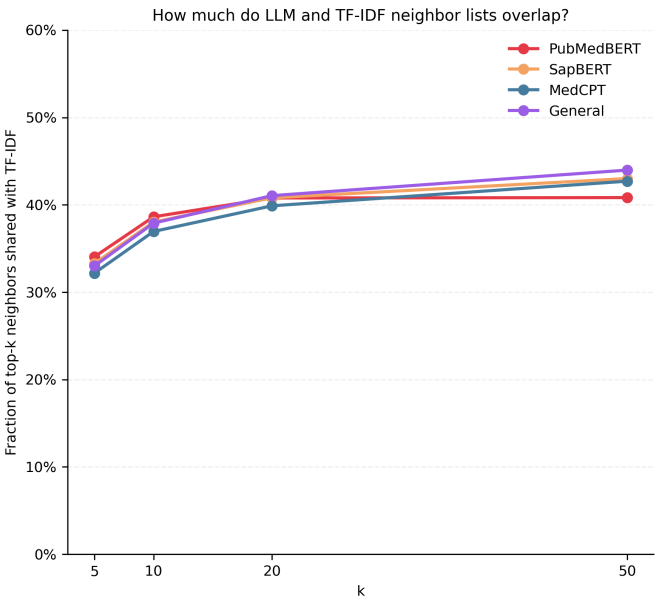


Fig. 4. Fraction of top-k LLM nearest neighbors shared with TF-IDF. Roughly $\sim$33% of nearest neighbors are shared at $k = 5$ across all four models, indicating two-thirds of each LLM's local structure is distinct from TF-IDF.

C. Text Embeddings Share Local Structure with RP but Not Atacformer

Despite statistically significant global rank correlations, top-k neighborhood overlap between text models and Atacformer was universally low: at $k = 5$, full-text models shared only 2 to 3% of nearest neighbors with Atacformer (Fig. 7). To test whether this reflects a fundamental limitation of text embeddings or an Atacformer-specific property, we repeated the analysis using RP as the reference (Fig. 8; see also Fig. 9).

The same full-text models shared 20 to 23% of nearest neighbors with RP at $k = 5$, a seven-fold increase relative to Atacformer (PubMedBERT: 22.5%; SapBERT: 22.1%; General: 21.4%; MedCPT: 19.8%). This dissociation persisted across all values of k tested. Cell-type-name-only models showed intermediate overlap with RP ($\sim$7% at $k = 5$),

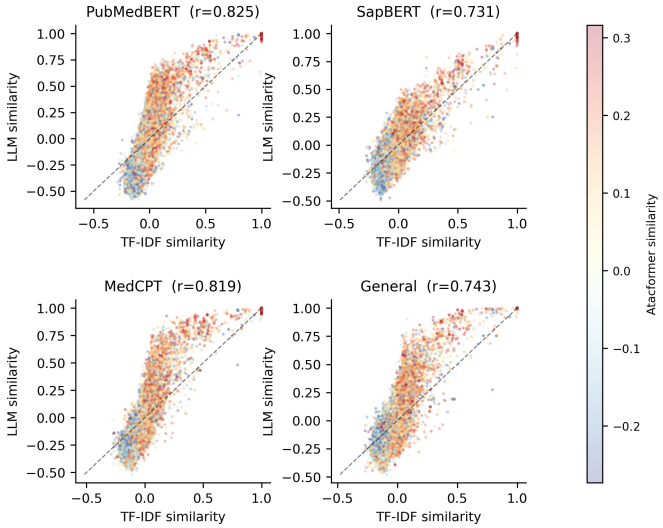


Fig. 5. TF-IDF (x-axis) vs. LLM (y-axis) pairwise similarity for 25,000 randomly sampled experiment pairs, colored by Atacformer similarity (RdYlBu; warmer = higher). Points above the diagonal indicate pairs the LLM scores higher than TF-IDF.

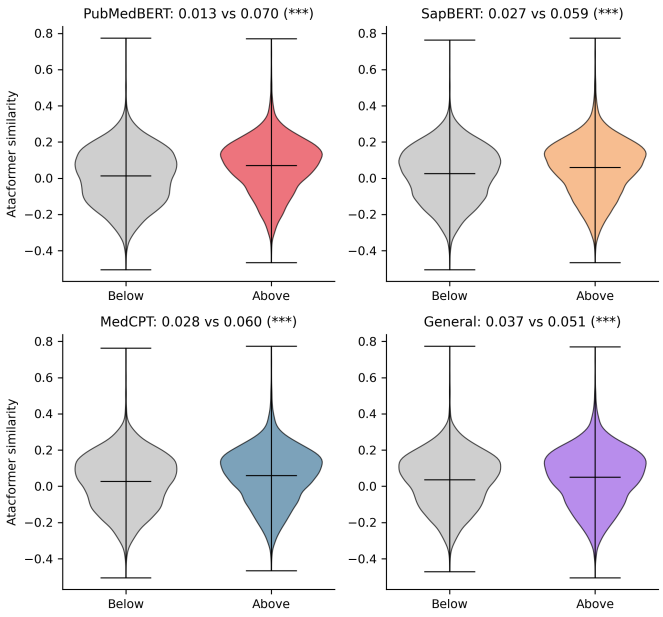


Fig. 6. Atacformer similarity for below- vs. above-diagonal pairs (Welch's t-test). For all four models, above-diagonal pairs have significantly higher mean Atacformer similarity (PubMedBERT: 0.070 vs. 0.013; SapBERT: 0.059 vs. 0.027; MedCPT: 0.060 vs. 0.028; General: 0.051 vs. 0.037); d denotes Cohen's d effect size (PubMedBERT: $d = 0.38$; SapBERT: $d = 0.22$; MedCPT: $d = 0.21$; General: $d = 0.09$).

substantially higher than their Atacformer overlap ($\sim$1%), consistent with RP partially encoding cell type identity through gene activation proximity.

We interpret this dissociation as follows. RP encodes a gene-activation-weighted summary of chromatin accessibility that closely tracks cell type identity. Text embeddings similarly cluster experiments by cell type through shared metadata vo-

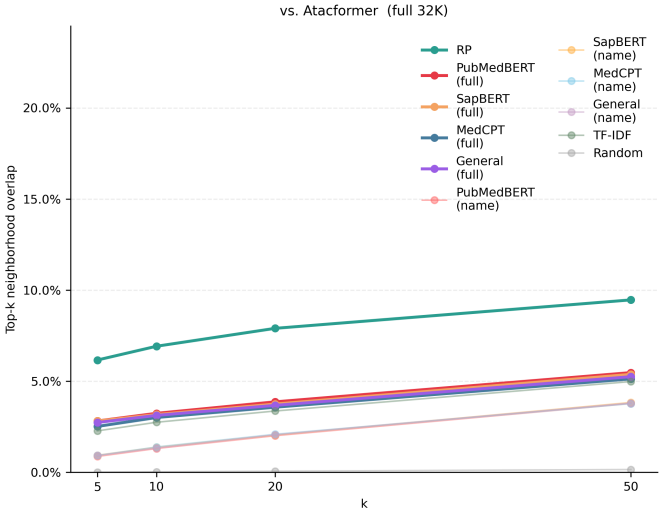


Fig. 7. Top-k neighborhood overlap vs. Atacformer at the full 32,245-experiment scale. Full-text models share only $\sim$2 to 3% of nearest neighbors with Atacformer at $k = 5$.

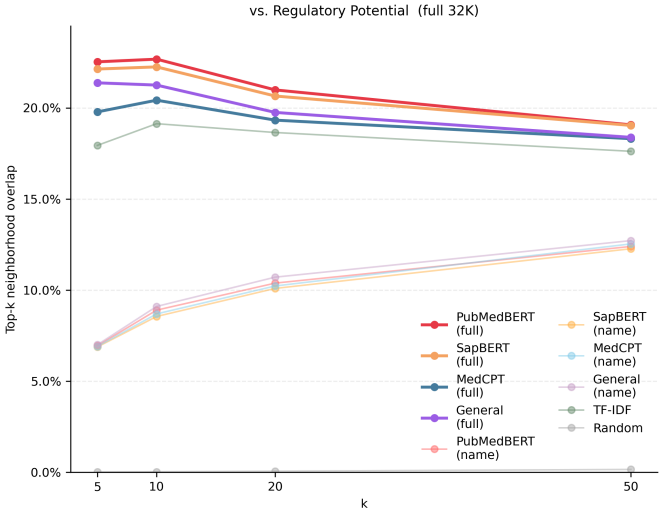


Fig. 8. Top-k neighborhood overlap vs. regulatory potential (RP) at the full 32,245-experiment scale. Full-text models share $\sim$20 to 23% of nearest neighbors with RP at $k = 5$, a seven-fold increase relative to Atacformer.

cabulary. Atacformer, by contrast, encodes the detailed peak-level chromatin accessibility landscape, capturing biological variation within and across cell types not reflected in text metadata. While Atacformer's broader similarity ordering partially correlates with text globally, its local neighborhoods are not predictable from metadata alone. These results show that text metadata is a viable proxy for cell-type-level clustering but not for the chromatin-specific similarity landscape encoded by Atacformer.

To test this directly, we computed the within-cell-type similarity for each embedding type: the mean pairwise cosine similarity among experiments sharing a cell-type label (947 cell types with $\geq$ 2 experiments). RP experiments of the same cell type are highly self-similar (median similarity 0.50),

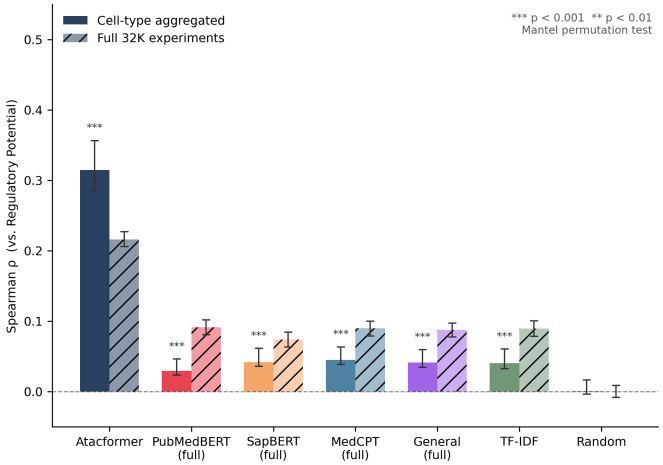


Fig. 9. Spearman rank correlation between each embedding type and regulatory potential (RP). Analogous to Table I with RP as reference (name-only variants omitted for clarity). Error bars show 95% bootstrap confidence intervals; significance markers (Mantel permutation test) are shown for the cell-type-aggregated bars only.

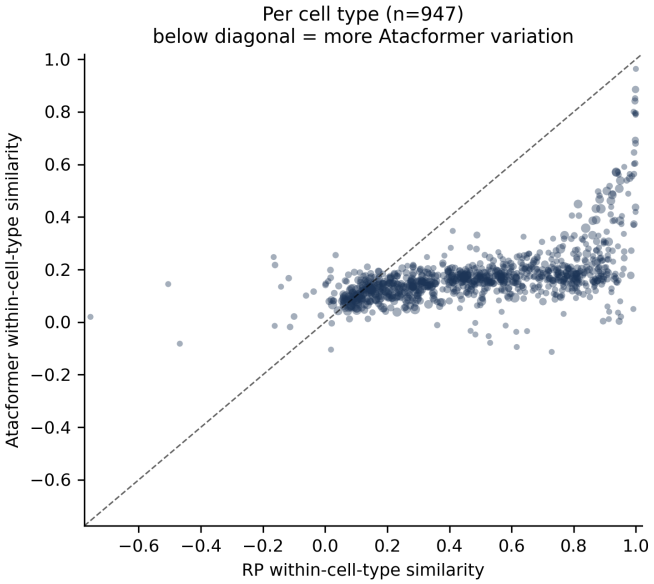


Fig. 10. Per-cell-type within-cell-type similarity for Atacformer vs. regulatory potential (RP), computed as mean pairwise cosine similarity among experiments sharing a cell-type label (947 cell types with ≥ 2 experiments). Points below the diagonal indicate higher within-cell-type variation for Atacformer than RP.

whereas Atacformer experiments of the same cell type are markedly less self-similar (median 0.16; Wilcoxon signed-rank $p = 9.1 \times 10^{-140}$), with Atacformer similarity lower than RP for 91.6% of cell types (Fig. 10). This confirms that RP representations are dominated by cell-type identity, while Atacformer retains substantial within-cell-type heterogeneity that text-derived metadata cannot resolve.

To illustrate this concretely, we examined pairs of experiments with high chromatin similarity (top 5%) but low TF-IDF similarity (below median; Tables II and III). For Atacformer,

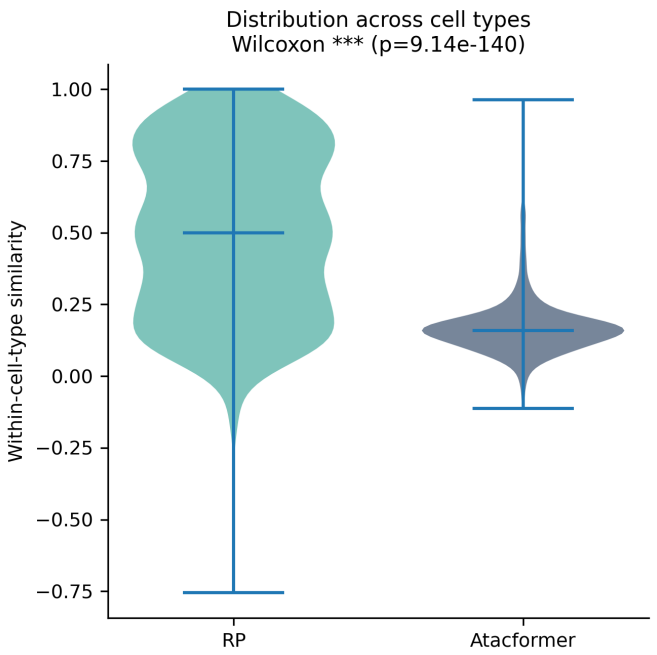


Fig. 11. Distribution of within-cell-type similarity across cell types (Wilcoxon signed-rank test). RP experiments are highly self-similar within cell type (median 0.50); Atacformer experiments are markedly less so (median 0.16), reflecting substantial within-cell-type heterogeneity.

these pairs link experiments with entirely different cell-type labels, e.g., the lung cancer line NCI-H1299 and CD4+ T cells (Atacformer similarity 0.37, TF-IDF similarity -0.20), or NCI-H1648 and induced pluripotent stem cell (iPSC)-derived neural cells (0.31 vs. -0.20). RP shows the same pattern at higher similarity values, e.g., NCI-H2347 and peripheral blood mononuclear cells (PBMC) (RP similarity 0.88, TF-IDF similarity -0.18). In each case, the text descriptions share essentially no vocabulary, yet the chromatin profiles are substantially more similar than a typical pair drawn from the dataset.

IV. Discussion

Our results establish quantitative bounds on the utility of experimental metadata as a proxy for ATAC-seq chromatin similarity, and show that language model pretraining provides limited and uneven value beyond simple vocabulary statistics.

The dominant finding is that vocabulary overlap (as measured by TF-IDF) accounts for the majority of the text-Atacformer correlation. TF-IDF achieves the highest unadjusted Spearman correlation at the full scale ($r = 0.300$) and fully explains the Atacformer alignment of both the general-purpose encoder and MedCPT. This suggests that experimental metadata primarily informs chromatin similarity through shared terminology, not semantic understanding.

The partial correlation analysis (Section 3.2) complicates this picture. PubMedBERT ($\Delta r = +0.051$) and SapBERT ($\Delta r = +0.045$) retain small but non-zero positive partial correlations with Atacformer after controlling for TF-IDF,

TABLE II
EXPERIMENT PAIRS: HIGH ATACFORMER SIMILARITY, LOW TF-IDF SIMILARITY. CELL-TYPE LABELS DIFFER ENTIRELY, ILLUSTRATING CHROMATIN SIMILARITY UNPREDICTABLE FROM METADATA.

Cell type A	Cell type B	Atac.	TFIDF
NCI-H1299	CD4+ T cells	0.374	−0.196
iPSC derived cells	PC-9	0.361	−0.176
NCI-H1648	iMeLCs	0.312	−0.214
(unlabeled)	CD4+ T cells	0.398	−0.147
NCI-H1648	iPSC neural cells	0.307	−0.204
(unlabeled)	IMR-90	0.309	−0.188
(unlabeled)	MV-4-11	0.331	−0.156
(unlabeled)	Neural crest cells	0.310	−0.179
iPSC derived cells	Jurkat	0.318	−0.164
(unlabeled)	DND-41	0.316	−0.161

TABLE III
EXPERIMENT PAIRS: HIGH RP SIMILARITY, LOW TF-IDF SIMILARITY, SELECTED ANALOGOUSLY TO TABLE II.

Cell type A	Cell type B	RP	TFIDF
iPSC derived cells	K-562	0.885	−0.221
NCI-H2347	PBMC	0.884	−0.181
RERF-LC-Ad1	XG-6	0.851	−0.224
(unlabeled)	PKC19	0.901	−0.171
iPSC derived cells	T-47D	0.874	−0.185
(unlabeled)	B-cell leukemia	0.891	−0.170
NCI-H1648	B cells	0.859	−0.181
NCI-H1703	CD4+ T cells	0.851	−0.186
NCI-H1299	CD4+ T cells	0.833	−0.196
(unlabeled)	ES cells	0.916	−0.154

whereas MedCPT ($\Delta r = -0.030$) and the general-purpose encoder ($\Delta r = -0.042$) do not. Since all three biomedical models share the same PubMedBERT-pretrained base, this divide is more plausibly attributable to fine-tuning objective than to pretraining. These partial correlations establish only that a non-zero residual exists, not that it constitutes biologically meaningful semantic understanding.

The global-versus-local dissociation between Atacformer and RP as reference structures reveals a complementary insight. Text embeddings capture cell-type-level clustering (which RP also encodes) but not the within-cell-type chromatin variation that Atacformer encodes. Text metadata may therefore support coarse-grained retrieval but should not substitute for chromatin similarity in fine-grained applications.

A central limitation is that Atacformer and RP are treated as proxies for true chromatin biology without independent validation against external ground truth. Both are transforms of the same ChIP-Atlas peak-call data and are therefore not independent reference structures. All conclusions are bounded by the assumption that these embeddings faithfully represent the chromatin biology they proxy; biases in either reference propagate directly to our text-chromatin agreement measurements.

Additional limitations include the heterogeneous quality of ChIP-Atlas experiments, the restriction to metadata fields available in ChIP-Atlas, and the absence of multimodal embeddings combining text and chromatin signals. Future work could explore whether fine-tuned text encoders trained to directly predict chromatin similarity, or retrieval-augmented multimodal approaches, could narrow the gap. The non-zero partial correlations for PubMedBERT and SapBERT suggest a residual signal that warrants further characterization, though establishing whether it reflects genuine biological semantics (rather than, e.g., shared structure with the reference embeddings themselves) would require validation against an independent biological ground truth.

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